

NEWCASTLE GRAMMAR SCHOOL



YEAR 12 2005 MATHEMATICS TRIAL EXAMINATION

*Time allowed – Three hours
(Plus 5 minutes reading time)*

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- ALL questions are of equal value.
- All necessary working should be shown in every question.
Marks may be deducted for careless or badly arranged work.
- Standard integrals are printed on page 8.
- Board-approved calculators may be used.
- Answer each question in a SEPARATE Writing Booklet.
- You may ask for extra Writing Booklets if you need them.

QUESTION 1 Use a SEPARATE Writing Booklet.**Marks**

- a) Calculate the value of $\sqrt[3]{2^{1.9}} + 7$ correct to four significant figures. 2
- b) Factorise completely $6x^3 - 48$ 2
- c) Express 300° in radians, in terms of π 2
- d) State the domain of the function with the equation $f(x) = \sqrt{3 - x}$ 2
- e) Differentiate $\frac{4}{x}$ 2
- f) Simplify completely $\log_a a^2 - \log_a \frac{1}{a}$ 2

QUESTION 2 Use a SEPARATE Writing Booklet.

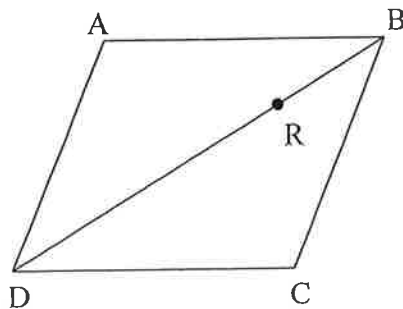
- a) Differentiate:
- i) $y = x^5 e^x$ 3
- ii) $y = \sqrt{\sin x}$ 3
- b) The points $A(-2,3)$, $B(3,8)$ and $C(10,9)$ are three of the vertices of parallelogram $ABCD$.
- i) Show that the coordinates of D are $(5,4)$. 2
- ii) Find the coordinates of the point K where the diagonals meet. 2
- iii) Prove that this parallelogram is a rhombus. 2

QUESTION 3 Use a SEPARATE Writing Booklet.**Marks**

- a) The probability that Mary-Anne catches no fish in any one day is 0.6. Find the probability that when Mary-Anne goes fishing for one week that she has at least one day where she does catch fish, answer correct to 2 decimal places. 2
- b) Find:
- i) $\int \sqrt{2x+1} \, dx$ 3
- ii) $\int \frac{2x}{4x^2+1} \, dx$ 3
- c) Solve $2 \sin^2 \theta + \sin \theta = 0$ for $0^\circ \leq \theta \leq 360^\circ$ 4

QUESTION 4 Use a SEPARATE Writing Booklet.

- a) For the function with the equation $y = 3 \sin 2x$:
- i) state the period of the function 1
- ii) state the amplitude of the function 1
- iii) sketch the graph of the function for $0 \leq x \leq 2\pi$ 2
- b) Find the values of k for which $x^2 - 2kx + 1 = 0$ has real roots 4
- c) ABCD is a rhombus. R is any point on diagonal BD.



- i) Prove that triangles ARD and CRD are congruent. 3
- ii) Hence, show that $AR = RC$. 1

QUESTION 5 Use a SEPARATE Writing Booklet.**Marks**

- a) For the parabola with the equation $y = x^2 - 4$ find the
- i) coordinates of the focus and the equation of the directrix. **4**
 - ii) exact volume generated when the area in the fourth quadrant, between the curve and the axes, is rotated around the y -axis. **3**
- b) Show, using calculus, that the graph with the equation $y = x^3(x - 4)$ has a horizontal inflection point at $x = 0$. **5**

QUESTION 6 Use a SEPARATE Writing Booklet.

- a) If α and β are the roots of the equation $3x^2 - 2x + 1 = 0$ find the value of $\alpha^2\beta + \alpha\beta^2$. **3**
- b) The area of an isosceles triangle, in which the two equal sides each have a length of 6 cm, is 10 cm^2 . Calculate the angle between the two equal sides, correct to the nearest minute. **4**
- c) i) Use the trapezoidal rule, with 4 sub-intervals (strips) to estimate the area between the curve $y = e^x$ and the x -axis, from $x = -1$ to $x = 1$, correct to 3 decimal places. **3**
- ii) Calculate the percentage error in this estimated area. **2**

QUESTION 7 Use a SEPARATE Writing Booklet.

Marks

- a) A loan of \$50 000 is taken out by a small business. The loan plus interest and charges are to be repaid at the end of each month in equal monthly instalments, \$M, over 8 years. Interest of 12% p.a. on the balance owing at the start of each month is added to the account at the end of each month. Additionally, at the end of each month a management charge of \$15 is added to the account. Let A_n be the amount owing after n months.

i) Show that $A_1 = 50000 \times 1.01 - (M - 15)$ 2

ii) Show that $A_2 = 50000 \times 1.01^2 - (M - 15)(1 + 1.01)$ 2

iii) Find the amount of each monthly instalment, \$M. 4

- b) A large tank of liquid chemical which contains L litres of chemical is being drained. The amount of chemical in the tank over time, t minutes, is given by:

$$L = 120(40 - t)^2$$

i) At what rate is the water draining out of the tank after 6 minutes? 2

ii) How long will it take for the tank to be completely empty? 2

QUESTION 8 Use a SEPARATE Writing Booklet.

- a) The number of bees, B , in a hive after t days is given by $B = 1200e^{kt}$, where k is a constant.

i) Show that $B = 1200e^{kt}$ satisfies $\frac{dB}{dt} = kB$ 2

ii) If there are 4 200 bees after 6 days, find the population after a further 10 days. 4

b) i) For $y = \frac{\log_e \sqrt{x}}{x}$ show that $\frac{dy}{dx} = \frac{1}{2} \left(\frac{1 - \log_e x}{x^2} \right)$ 3

ii) Hence, evaluate $\int_1^e \frac{1 - \log_e x}{x^2} dx$, answer in exact form. 3

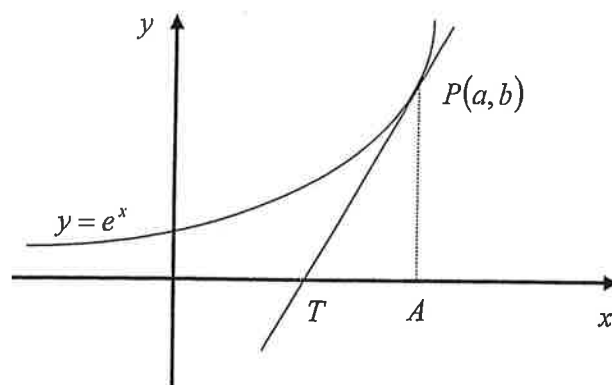
QUESTION 9 Use a SEPARATE Writing Booklet.

Marks

a) Solve $2 \log_e x = \log_e (x + 6)$

3

b) The point $P(a, b)$ lies on the curve with the equation $y = e^x$.



i) Show that the tangent at P has the equation

3

$$y - e^a = e^a(x - a)$$

ii) Hence, find the coordinates of T , the point where the tangent meets the x -axis. Give your answer in terms of a .

2

iii) A is the foot of the perpendicular from P to the x -axis. Show that the length of the interval TA is constant, for any point P .

2

c) Find $\int \tan x \, dx$

2

QUESTION 10 Use a SEPARATE Writing Booklet.

Marks

Two particles A and B move along the x -axis, both starting when $t = 0$. The displacement of particle A is given by $x = 4t + 21 - t^2$, where x is measured in metres and t is in seconds. The displacement of particle B is given by $x = 2t(t - 7)$.

- | | | |
|------|--|---|
| i) | Find when and where particle A is stationary. | 2 |
| ii) | On the same diagram, sketch each particle's displacement graph, showing <u>only</u> the t -intercepts. | 3 |
| iii) | Show that the particles meet <u>only</u> after 7 seconds. | 1 |
| iv) | Show that the distance, D , between the two particles at any time, t , is given by $D = 18t + 21 - 3t^2$ | 1 |
| v) | During the first 7 seconds, when are the particles furthest apart? | 3 |
| vi) | Find the time when both particles have the same velocity. | 2 |

1) $\sqrt[3]{2^{1.9} + 7}$: on calculator :-

$$\sqrt[3]{(2 \wedge 1.9 + 7)} =$$

$$= \underbrace{2.20577 \dots}_{4 \text{ sig. fig.}}$$

$$= \boxed{2.206}$$

b) $6x^3 - 48 = 6(x^3 - 8)$

$$= 6((x)^3 - (2)^3)$$

using: $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$

$$\therefore \boxed{= 6(x-2)(x^2 + 2x + 4)}$$

c) $\frac{300^\circ}{1} \times \frac{\pi}{180} = \boxed{\frac{5\pi}{3} \text{ rad}}$

d) For domain: we need:-

$$3 - x \geq 0$$

$$\text{i.e. } \boxed{3 \geq x \text{ or } x \leq 3}$$

e) $\frac{d}{dx}\left(\frac{4}{x}\right) = 4 \times \frac{d}{dx}\left(\frac{1}{x}\right)$
 $= 4 \times \log_e x$
 $= \boxed{4 \log_e x \text{ (or } 4 \ln x)}$

f) $\log_a a^2 - \log_a \frac{1}{a} = \log_a \left(\frac{a^2}{1/a}\right)$
 $= \log_a (a^2 \div a^{-1})$
 $= \log_a (a^3)$
 $= \underline{\underline{3}}$

2) $y = x^5 e^x$ $u = x^5$ $v = e^x$
a) $\therefore u' = 5x^4$ $\therefore v' = e^x$

$$y' = vu' + uv'$$

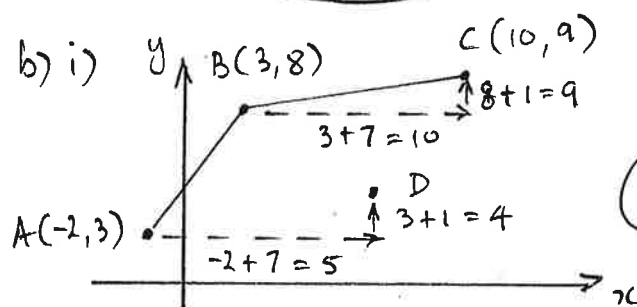
$$= e^x \times 5x^4 + x^5 \times e^x$$
 ③

$$= \boxed{x^4 e^x (5 + x)}$$

b) $y = \sqrt{\sin x}$
 $= (\sin x)^{\frac{1}{2}}$ ③

$$\therefore \frac{dy}{dx} = \frac{1}{2} (\sin x)^{-\frac{1}{2}} \times \cos x$$

$$= \boxed{\frac{\cos x}{2\sqrt{\sin x}}}$$



$$\therefore D(5, 4) \text{ (QED)}$$

ii) Diagonals of parallelogram BISECT
 $\therefore$ find midpoint of AC (or BD)

$$\therefore K = \left(\frac{-2+10}{2}, \frac{3+9}{2}\right)$$
 ②

$$= \boxed{(4, 6)}$$

iii) Parallelogram = rhombus
if adjacent pair of sides equal
 $\therefore$ show $AB = BC$ ②
 $AB = \sqrt{(3+2)^2 + (8-3)^2} = \sqrt{50} \text{ units}$
 $BC = \sqrt{(10-3)^2 + (9-8)^2} = \sqrt{50} \text{ units}$
 $\therefore AB = BC \therefore \text{rhombus (QED)}$

③ a) $P(\text{at least 1 day catch fish})$

$$= 1 - P(\text{no fish})$$

$$= 1 - (0.6)^7 \leftarrow \text{for 7 days}$$

$$= 0.972 \dots$$

$$= \boxed{0.97}$$

b) i) $\int \sqrt{2x+1} \, dx$

$$= \int (2x+1)^{\frac{1}{2}} \, dx$$

$$= \frac{(2x+1)^{\frac{3}{2}}}{\frac{3}{2} \times 2} + C$$

using: $\int (ax+b)^n \, dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$

$$= \frac{(2x+1)^{\frac{3}{2}}}{3} + C$$

$$\text{or } \frac{1}{3} \sqrt{(2x+1)^3} + C$$

ii) $\int \frac{2x}{4x^2+1} \, dx$

$$= \frac{1}{4} \int \frac{8x}{4x^2+1} \, dx$$

$$= \boxed{\frac{1}{4} \log_e (4x^2+1) + C}$$

{ using: $\int \frac{f'(x)}{f(x)} \, dx = \log_e(f(x)) + C$ }

(2)

c) $2 \sin^2 \theta + \sin \theta = 0$

$$\therefore \sin \theta (2 \sin \theta + 1) = 0$$

$$\downarrow \qquad \qquad \downarrow$$

$$\therefore \sin \theta = 0 \quad \text{or} \quad 2 \sin \theta + 1 = 0$$

boundary angle
or

use sine graph
 π -intercepts

$$\therefore \sin \theta = -\frac{1}{2}$$

$$\text{Gap } L = 30^\circ$$

$\sin \theta < 0 \therefore$ 3rd/4th quadrants

$$\therefore \theta = 0^\circ, 180^\circ$$

$$\text{or } 360^\circ$$

$$\therefore \theta = 180^\circ + 30^\circ,$$

$$360^\circ - 30^\circ$$

$$\therefore \theta = 210^\circ, 330^\circ$$

$$\therefore \boxed{\theta = 0^\circ, 180^\circ, 210^\circ, 330^\circ, 360^\circ}$$

④ a) $y = 3 \sin 2x$

i) period: $P = \frac{2\pi}{n}$

$$= \frac{2\pi}{2}$$

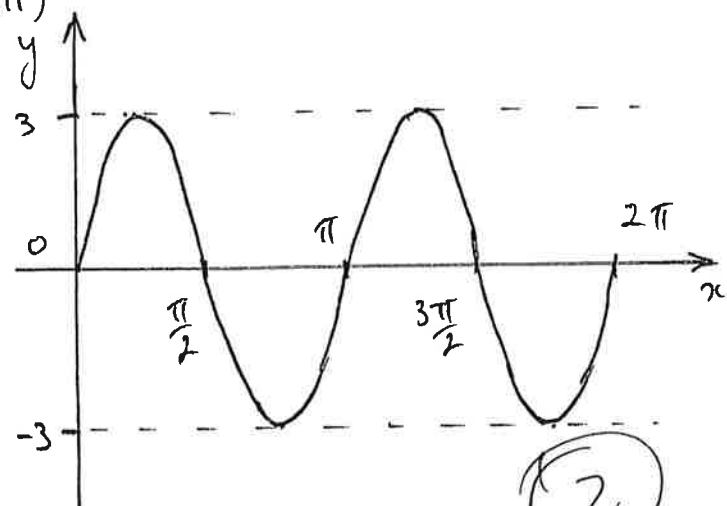
$$\therefore \boxed{P = \pi}$$

(1)

ii) amplitude = $\boxed{3}$

(1)

iii)



(2)

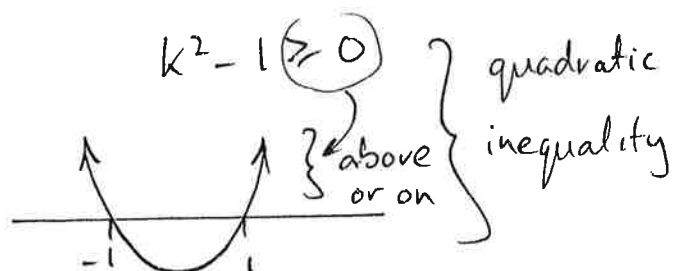
④ b) For real roots we need:

$$\Delta \geq 0$$

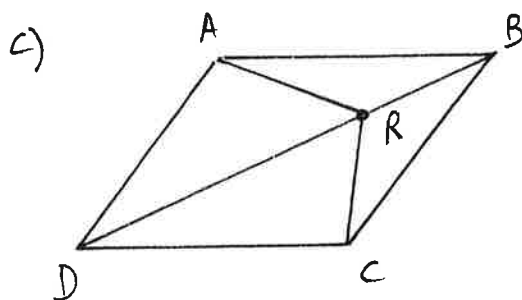
ie. $b^2 - 4ac \geq 0$

$$\therefore (-2k)^2 - 4 \times 1 \times 1 \geq 0$$

$$\therefore 4k^2 - 4 \geq 0 \quad (\div 4) \quad \textcircled{4}$$



$$\therefore k \leq -1 \text{ or } k \geq 1$$



i) $AD = CD$ (sides of rhombus) S
 $\angle ADR = \angle CDR$ (L's bisected by diagonal) A

③ $DR = DR$ (common) S

$\therefore \triangle ARD \equiv \triangle CRD$ (SAS)

ii) $AR = RC$

① corresponding sides
 in cong. Δ 's

⑤ a) $y = x^2 - 4$ } down 4 $\otimes$

i) $\therefore x^2 = y + 4$

or: $(x-0)^2 = 1(y+4)$

ie. $(x-0)^2 = 4 \times \frac{1}{4} (y - (-4))$

FORMULA: $(x-h)^2 = 4a(y-k)$

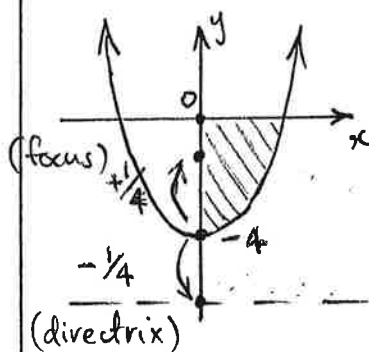
Vertex = (h, k)

Focal length = a

$\therefore$ Vertex at $(0, -4)$ ($\otimes$ as above)

Focal length = $\frac{1}{4}$

$\therefore$ Sketch:



$\therefore$ Focus is:

$$(0, -3\frac{3}{4})$$

Directrix is:-

$$y = -4\frac{1}{4}$$

ii) $V = \pi \int_a^b x^2 dy$ } revolving (around) y-axis

$\therefore V = \pi \int_{-4}^0 y+4 dy$ } of shaded area in diagram above

$$= \pi \left[\frac{y^2}{2} + 4y \right]_{-4}^0$$

$$= \pi \left[\left(\frac{0^2}{2} + 4 \times 0 \right) - \left(\frac{(-4)^2}{2} + 4 \times (-4) \right) \right]$$

$$= \pi [(0) - 8 + 16]$$

$$= \textcircled{8\pi \text{ units}^3}$$

⑤ b) Horizontal $\Rightarrow \frac{dy}{dx} = 0 \dots (1)$

Inflection $\Rightarrow \frac{d^2y}{dx^2} = 0 \dots (2)$
 AND sign changes

(1) $y = x^3(x-4)$
 $= x^4 - 4x^3$

$\therefore \frac{dy}{dx} = 4x^3 - 12x^2$
 $= x^2(4x - 12)$

Letting: $x^2(4x - 12) = 0$
 $\downarrow \qquad \qquad \downarrow$
 $x^2 = 0 \qquad \text{not required}$
 $\therefore x = 0$

$\therefore$ St. pt. at $x=0$
 (i.e. horizontal)

(2) $\frac{d^2y}{dx^2} = 12x^2 - 24x$
 $= 12x(x-2)$

Letting: $12x(x-2) = 0$
 $\downarrow$
 $12x = 0$
 $\therefore x = 0$

$\therefore$ POSSIBLE inf. pt. at $x=0$.

AND

x	-1	0	1
$\frac{d^2y}{dx^2}$	+	0	-

at $x = -1$: $\frac{d^2y}{dx^2} = 12(-1)^2 - 24(-1) = +36$

$x = 1$: $\frac{d^2y}{dx^2} = 12(1)^2 - 24(1) = -12$

in sign change: (inf. pt. at $x=0$)

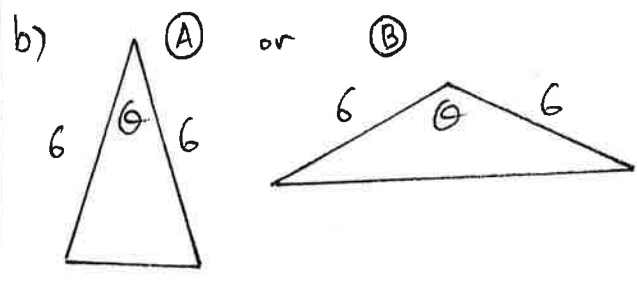
⑥ a) $3x^2 - 2x + 1 = 0$

$\therefore \alpha + \beta = -b/a$
 $= -(-2)/3$
 $= 2/3$

$\alpha\beta = c/a$
 $= 1/3$

③

and $\alpha^2\beta + \alpha\beta^2 = \alpha\beta(\alpha + \beta)$
 $= \frac{1}{3} \times \frac{2}{3}$
 $= \frac{2}{9}$



Area = $\frac{1}{2}ab \sin C$

$\therefore \frac{1}{2} \times 6 \times 6 \times \sin \theta = 10$

$18 \times \sin \theta = 10$

$\sin \theta = 10/18$

$\therefore \theta = \sin^{-1}(5/9)$

$= 33^\circ 44' 56.36'' \dots$

$\therefore \theta = 33^\circ 45'$ (for A)

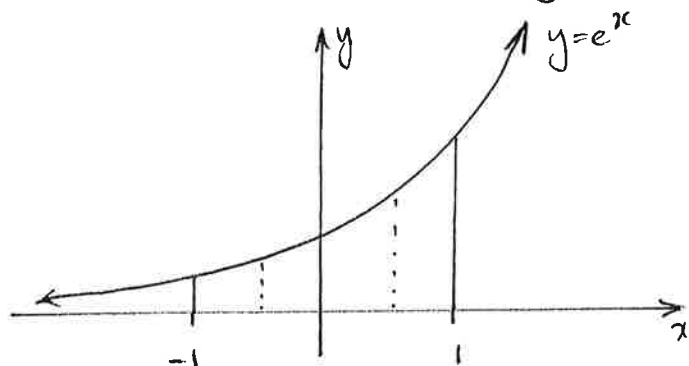
OR $\theta = 180^\circ - 33^\circ 45'$

ie $\theta = 146^\circ 15'$ (for B)

(i.e. two possible answers, for θ acute or obtuse)

4

c) i) Sketch (not necessary)



Area:

x	y	x	
-1	0.3679	1	0.3679
-0.5	0.6065	2	1.2131
0	1	2	2
0.5	1.6487	2	3.2974
1	2.7183	1	2.7183

TOTAL = 9.5967

$$\therefore \text{Area} \div \frac{h}{2} \times \text{TOTAL}$$

$$= \frac{0.5}{2} \times 9.5967$$

$$= 2.3991 \dots$$

$$\therefore \text{Area} = 2.399 \text{ units}^2 \quad (3 \text{ dp})$$

ii) Exact Area = $\int_{-1}^1 e^x dx$

$$= [e^x]_{-1}^1$$

$$= e^1 - e^{-1}$$

$$= 2.350 \text{ units}^2 \quad (3 \text{ dp})$$

$$\therefore \text{Percentage error} = \frac{2.399 - 2.350}{2.350} \times \frac{100}{1}$$

$$= \frac{0.049}{2.35} \times 100$$

$$= 2.08 \dots$$

$$\therefore \% \text{ error} = 2.1\% \quad (1 \text{ dp})$$

⑦ a) i) $A_1 = \$50000 + 1\% \text{ per month interest}$
 $+ \$15 - M$

$$= 50000 \times 1.01 - M + 15$$

$$= 50000 \times 1.01 - (M - 15) \text{ (RED)}$$

ii) $\therefore A_2 = A_1 \times 1.01 - (M - 15)$

$$= \{50000 \times 1.01 - (M - 15)\} \times 1.01 - (M - 15)$$

$$= 50000 \times 1.01^2 - 1.01(M - 15) - (M - 15)$$

$$= 50000 \times 1.01^2 - (M - 15)(1 + 1.01) \text{ (RED)}$$

iii) From ii) we "generalise":

$$\therefore A_{96} = 50000 \times 1.01^{96} - (M - 15)(1 + 1.01 + \dots + 1.01^{95})$$

G.P. $a = 1, r = 1.01$
 $n = 96$

$$\therefore S_n = \frac{a(r^n - 1)}{r - 1} = \frac{1(1.01^{96} - 1)}{1.01 - 1}$$

AND $A_{96} = 0$
 for loan fully repaid

$$\therefore 0 = 50000 \times 1.01^{96} - (M - 15) \frac{(1.01^{96} - 1)}{0.01}$$

$$\therefore M = \left\{ 50000 \times 1.01^{96} \times \frac{0.01}{1.01^{96} - 1} \right\} + 15$$

$$= 827.642 \dots$$

$$\therefore M = \$827.64$$

⑦ b) $L = 120(40-t)^2$

i) $\therefore \text{Rate} = \frac{dL}{dt} = 120 \times 2(40-t) \times -1$
 $= -240(40-t)$

$\therefore$ at $t = 6$:

$$\frac{dL}{dt} = -240(40-6)$$

$$= \boxed{-8160 \text{ l/min}}$$

ii) For empty tank: $L = 0$

i.e. $120(40-t)^2 = 0$

$$\therefore (40-t)^2 = 0$$

$$\therefore 40-t = 0$$

$$\therefore t = 40 \text{ minutes}$$

⑧ a) i) $B = 1200e^{kt}$

② $\therefore \frac{dB}{dt} = 1200 \times k e^{kt}$
 $= k(1200e^{kt})$ from above
 i.e. $\frac{dB}{dt} = kB$ (QED)

ii) $4200 = 1200e^{6k} \quad \div 1200$

$$e^{6k} = 7/2$$

④ $\log_e(e^{6k}) = \log_e(7/2)$

$$\therefore 6k = \log_e(7/2)$$

$$k = \log_e(7/2) \div 6$$

$$= 0.1088 \text{ (4dp)}$$

$\therefore$ at $t = 16$ days

$$B = 1200e^{0.1088 \times 16}$$

$$= \boxed{33890} \text{ (nearest whole)}$$

b) i) $y = \frac{\log_e(x^{1/2})}{x}$

$$= \frac{1/2 (\log_e x)}{x}$$

③ $= \frac{1}{2} \left(\frac{\log_e x}{x} \right)$ $u = \log_e x, v = x$
 $u' = \frac{1}{x}, v' = 1$

$$\therefore \frac{dy}{dx} = \frac{1}{2} \left(\frac{vu' - uv'}{v^2} \right)$$

$$= \frac{1}{2} \left(\frac{x \times \frac{1}{x} - \log_e x \times 1}{x^2} \right)$$

$$= \boxed{\frac{1}{2} \left(\frac{1 - \log_e x}{x^2} \right)} \text{ (QED)}$$

ii) $\therefore \int_1^e \frac{1 - \log_e x}{x^2} dx$

$$= 2 \int_1^e \frac{1}{2} \left(\frac{1 - \log_e x}{x^2} \right) dx$$

$$= 2 \left[\frac{\log_e \sqrt{x}}{x} \right]_1^e$$

$$= 2 \left\{ \left(\frac{\log_e e^{1/2}}{e} \right) - \left(\frac{\log_e \sqrt{1}}{1} \right) \right\}$$

$$= 2 \left\{ \frac{1/2}{e} - \frac{0}{1} \right\}$$

$$= 2 \times \frac{1/2}{e}$$

$$= \boxed{\frac{1}{e}}$$

⑨ a) $2 \log_e x = \log_e (x+6)$

$$\therefore \cancel{\log_e (x^2)} = \cancel{\log_e (x+6)}$$

$$\therefore x^2 = x+6$$

$$x^2 - x - 6 = 0$$

$$(x-3)(x+2) = 0$$

$$\therefore x = -2 \text{ or } 3$$

BUT by substitution:

at $x = -2$: $2 \log_e (-2) = \log_e (-2+6)$

i.e. $2 \log_e (-2) = \log_e (4)$

$\log_e x$ undefined
for negative x

$\therefore$ ONLY solution is: $x = 3$

b) i) $y - y_1 = m(x - x_1)$

at $x_1 = a$, $y_1 = e^a$

and $\frac{dy}{dx} = e^x \therefore m = e^a$

$$\therefore y - e^a = e^a (x - a) \text{ (QED)}$$

ii) For T: let $y = 0$:-

i.e. $0 - e^a = e^a (x - a) \quad (\div e^a)$

$$-1 = x - a$$

$$\therefore x = a - 1$$

iii) A + T: $x = a - 1$

A + A: $x = a$

$$\therefore TA = a - (a - 1)$$

$$= \cancel{a} - \cancel{a} + 1$$

$$\therefore TA = 1 \text{ unit} \quad (\text{QED})$$

ie constant for all P

(as independent of 'a')

c) $\int \tan x \, dx$

$$= \int \frac{\sin x}{\cos x} \, dx$$

$$= - \int \frac{-\sin x}{\cos x} \, dx$$

$$= - [\log_e (\cos x)] + C$$

or $= - \log_e (\cos x) + C$

⑩ i) A: $x = 4t + 21 - t^2$

$$\therefore \frac{dx}{dt} = v_A = 4 - 2t$$

$\therefore$ stationary when $v_A = 0$

i.e. $4 - 2t = 0$

$$2t = 4$$

$$t = 2 \text{ seconds}$$

and at $t = 2$: $x = 4 \times 2 + 21 - 2^2$

$$x = 25 \text{ m}$$

②

(10) ii) (A) $x = 4t + 21 - t^2$ } concave down
 $= -(t^2 - 4t - 21)$
 $= -(t-7)(t+3)$

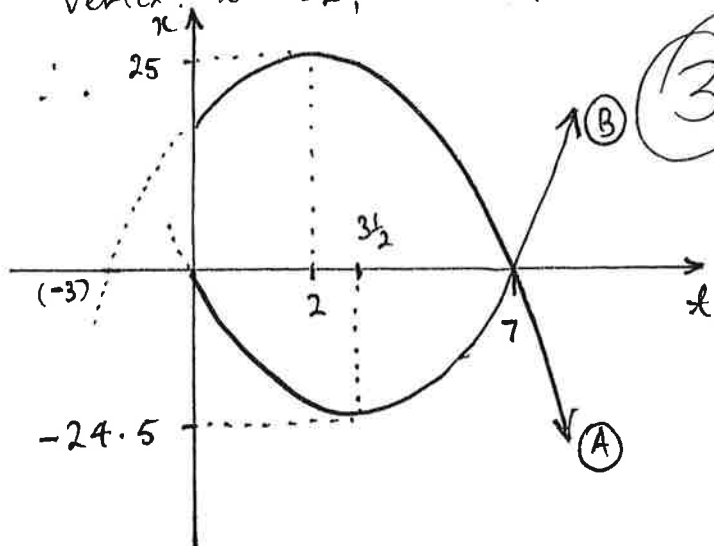
$\therefore t$ -int's: $t = -3$ or 7

Vertex: $t = 2$, $x = 25$ (from i)

(B) $x = 2t(t-7)$

$\therefore t$ -int's: $t = 0$, $t = 7$

Vertex: $t = 3\frac{1}{2}$, $x = -24.5$



iii) From graph:

Particles meet when x -values equal

ie. graphs intersect

We see graphs intersect only once at $t = 7$ (QED)

(OR) Algebra: $4t + 21 - t^2 = 2t(t-7)$
 $= 2t^2 - 14t$

$\therefore 3t^2 - 18t - 21 = 0$ ($\div 3$)

$t^2 - 6t - 7 = 0$

$(t-7)(t+1) = 0$

$\therefore t = -1$ (not possible)

iv) Distance between particles: D

$D = (A) - (B)$ } for $t=0$ to $t=7$
 $= 4t + 21 - t^2 - (2t^2 - 14t)$
 $= 4t + 21 - t^2 - 2t^2 + 14t$

$\therefore D = 18t + 21 - 3t^2$ (QED)

v) $D_{\max}$ at $\frac{dD}{dt} = 0$

ie $18 - 6t = 0$

$\therefore 6t = 18$

$\therefore t = 3$ seconds

NOTE $\frac{d^2D}{dt^2} = -6 < 0$ at $t = 3$

$\therefore t = 3$ seconds is a maximum

vi) Same velocity at $v_A = v_B$

and $x_B = 2t^2 - 14t$

$\therefore v_B = 4t - 14$

$\therefore$ need: $4 - 2t = 4t - 14$

$18 = 6t$

$\therefore t = 3$ seconds

for same velocities